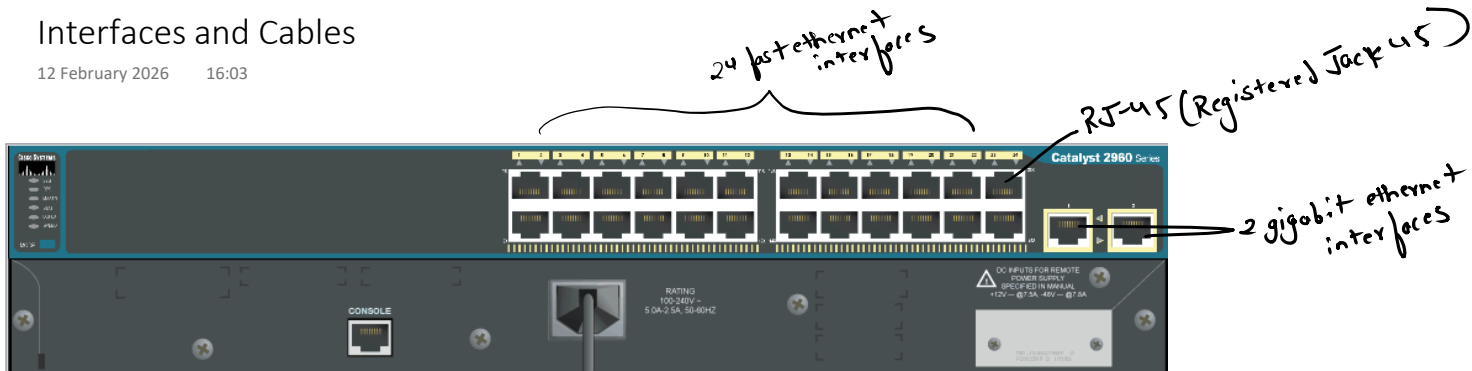


Interfaces and Cables

12 February 2026 16:03



CISCO CATALYST 2060 SERIES

RJ-45

- RJ-45 is a connector used at the ends of Ethernet copper cables (like Cat5e, Cat6).

What is Ethernet?

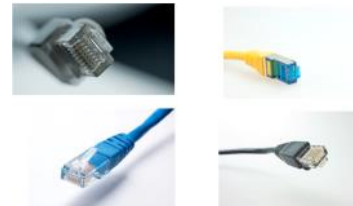
- Ethernet is a collection of networking standards and protocols used to connect devices in a local area network (LAN).

What is a Protocol?

- A protocol is a standardized set of rules that define how data is formatted, transmitted, and processed between devices.

Why do we need Network Protocols?

- To provide clear rules and standards for communication.
- To ensure interoperability between different hardware and software.
- To enable reliable and secure data transfer (e.g., encryption).
- To support smooth and scalable network growth.

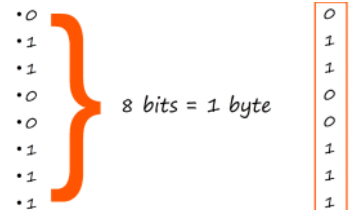


RJ-45 (Registered Jacket)

Bits and Bytes

Communication between two or more devices happens in **bits**. That's why network speed is measured in **bits per second (bps)**.

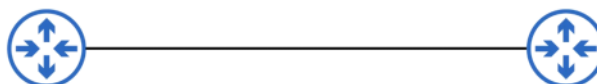
Bits flow **one by one**, not as groups or packets of bytes all at once. Each bit travels sequentially across the communication medium.



Speed is measured in bits per second (Kbps, Mbps, Gbps, etc), not bytes per second

- 1 kilobit (Kb) = 1,000 bits
- 1 megabit (Mb) = 1,000,000 bits
- 1 gigabit (Gb) = 1,000,000,000 bits
- 1 terabit (Tb) = 1,000,000,000,000 bits

Decimal		
Value	SI	
1000 10 ³	kbit	kilobit
1000 ² 10 ⁶	Mbit	megabit
1000 ³ 10 ⁹	Gbit	gigabit
1000 ⁴ 10 ¹²	Tbit	terabit
1000 ⁵ 10 ¹⁵	Pbit	petabit
1000 ⁶ 10 ¹⁸	Ebit	exabit
1000 ⁷ 10 ²¹	Zbit	zettabit
1000 ⁸ 10 ²⁴	Ybit	yottabit

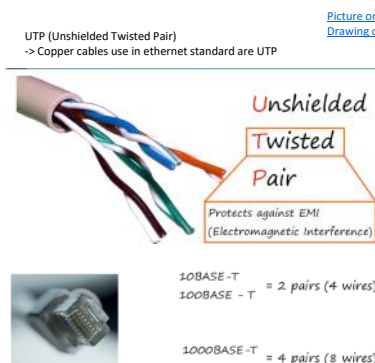


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Ethernet standards are defined in IEEE 802.3 standard in 1983
IEE - Institute of electrical and electronics engineers

Speed	Common Name	IEEE Standard	Informal Name	Maximum Length
10 Mbps	Ethernet	802.3i	10BASE-T	100 m
100 Mbps	Fast Ethernet	802.3u	100BASE-T	100 m
1 Gbps	Gigabit Ethernet	802.3ab	1000BASE-T	100 m
10 Gbps	10 Gig Ethernet	802.3an	10GBASE-T	100 m

Base = Baseband signal
T - Twisted Pair

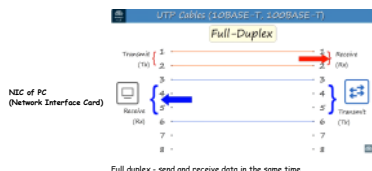


Not all ethernet cables uses 8 pairs

<https://www.ieee802.org/3/>

[illegible]

Straight Through Cable VS Crossover Cable



Straight-Through Cable

A straight-through cable is a type of Ethernet cable where both ends are wired using the same standard:

- T568A to T568A, or
- T568B to T568B

This means the wire color order is identical on both ends of the cable.

Wiring Standard

- One end: T568A
- Other end: T568A

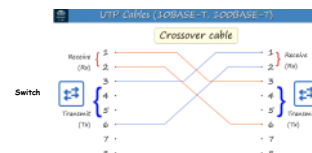
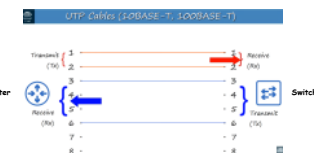
OR

- One end: T568B
- Other end: T568B

Used To Connect Different Devices

A straight-through cable is commonly used to connect different types of devices, such as:

- Switch to Router
- Router to Switch
- PC to Switch
- PC to Router



Crossover Cable

A crossover cable is an Ethernet cable where both ends are wired using different standards.

Wiring Standard

- One end: T568A
- Other end: T568B

OR

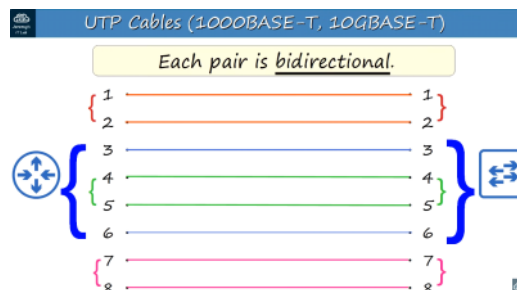
- One end: T568B
- Other end: T568A

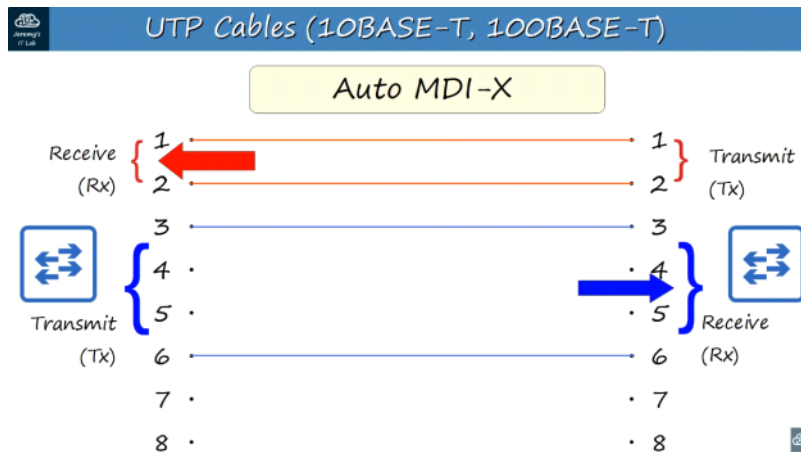
This type of cable is typically used to connect similar devices, such as:

- PC to PC
- Switch to Switch
- Router to Router

https://www.reddit.com/r/ccna/comments/3qjzyl/pc_help_with_a_crossover_cable/

UTP Cables (10BASE-T, 100BASE-T)			
Device Type		Transmit (Tx) Pins	Receive (Rx) Pins
Router		1 and 2	3 and 6
Firewall		1 and 2	3 and 6
PC		1 and 2	3 and 6
Switch		3 and 6	1 and 2





What is MDI and MDI-X?

In Ethernet, devices use specific pins for transmit (Tx) and receive (Rx).

- **MDI (Medium Dependent Interface)**
 - Used by: **PCs, routers**
 - Transmit: pins 1 & 2
 - Receive: pins 3 & 6
- **MDI-X (Medium Dependent Interface Crossover)**
 - Used by: **Switches, hubs**
 - Receive: pins 1 & 2
 - Transmit: pins 3 & 6

Because of this difference:

- **Different device types** (MDI ↔ MDI-X) → use **straight-through cable**
- **Same device types** (MDI ↔ MDI or MDI-X ↔ MDI-X) → use **crossover cable**

What is Auto-MDI/MDI-X?

Auto-MDI/MDI-X is a feature in modern network devices that:

- Automatically detects the type of device connected.
- Automatically switches its Tx and Rx pins if needed.
- Eliminates the need to choose between straight-through and crossover cables.

So:

- You can use **any Ethernet cable**.
- The device adjusts internally.

How it works (simple steps)

1. Device detects a cable connection.
2. It sends test signals.
3. It checks whether it is receiving signals on expected pins.
4. If not, it automatically swaps Tx and Rx internally.
5. Communication starts normally.

Example

Without Auto-MDI-X:

- PC → PC requires **crossover cable**.

With Auto-MDI-X:

- PC → PC works with **straight-through or crossover**.

Fiber-Optic Connections



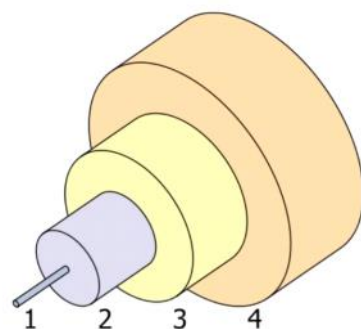
The SFP transceiver is inserted into the switch port, and the fiber optic cable is connected to the SFP module.



SFP Transceiver
(Small Form-Factor Pluggable)



Fiber-Optic Connections



Original by Bob Mellish, SVG derivative by Benchill
(https://commons.wikimedia.org/wiki/File:Singlemode_fibre_structure.svg), "Singlemode fibre structure", <https://commons.wikimedia.org/licenses/by-sa/3.0/legalcode>

- 1: the fiberglass core itself
- 2: cladding that reflects light
- 3: a protective buffer
- 4: the outer jacket of the cable

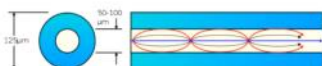
single-mode

multimode

Single-Mode Fiber

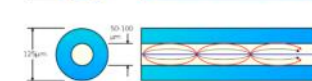
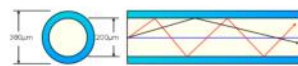


Mirzaei (https://commons.wikimedia.org/wiki/File:Optical_fiber_types.svg), "Optical fiber types", edited, <https://creativecommons.org/licenses/by-sa/3.0/legalcode>



- Core diameter is narrower than multimode fiber.
- Light enters at a single angle (mode) from a laser-based transmitter.
- Allows longer cables than both UTP and multimode fiber.
- More expensive than multimode fiber (due to more expensive laser-based SFP transmitters)

Multimode Fiber



Mirzaei (https://commons.wikimedia.org/wiki/File:Optical_fiber_types.svg), "Optical fiber types", edited, <https://creativecommons.org/licenses/by-sa/3.0/legalcode>

- Core diameter is wider than single-mode fiber.
- Allows multiple angles (modes) of light waves to enter the fiberglass core.
- Allows longer cables than UTP, but shorter cables than single-mode fiber.
- Cheaper than single-mode fiber (due to cheaper LED-based SFP transmitters).

Fiber-Optic Cable Standards

Informal Name	IEEE Standard	Speed	Cable Type	Maximum Length
1000BASE-LX	802.3z	1 Gbps	Multimode or Single-Mode	550 m (MM) 5 km (SM)
10GBASE-SR	802.3ae	10 Gbps	Multimode	400 m
10GBASE-LR	802.3ae	10 Gbps	Single-Mode	10 km
10GBASE-ER	802.3ae	10 Gbps	Single-Mode	30 km



UTP vs Fiber-Optic Cabling

UTP

- Lower cost than fiber-optic.
- Shorter maximum distance than fiber-optic (~100m).
- Can be vulnerable to EMI (Electromagnetic Interference).
- RJ45 ports used with UTP are cheaper than SFP ports.
- Emit (leak) a faint signal outside of the cable, which can be copied (=security risk)

Fiber-Optic

- Higher cost than UTP.
- Longer maximum distance than UTP.
- No vulnerability to EMI.
- SFP ports are more expensive than RJ45 ports (single-mode is more expensive than multimode).
- Does not emit any signal outside of the cable (=no security risk).

MCQs

13 February 2026 01:15

You connect two old routers together with a UTP cable, however data is not successfully sent and received between them. What could be the problem?

- a) They are connected with a straight-through cable.*
- b) They are connected with a crossover cable.*
- c) They are operating in Auto MDI-X mode.*

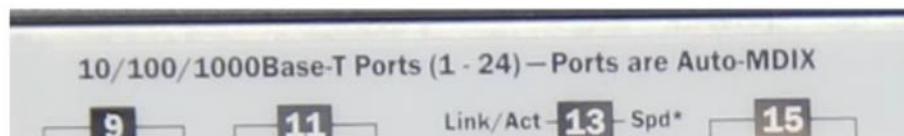
Your company wants to connect switches in two separate buildings that are about 150 meters apart. They want to keep costs down, if possible. What kind of cable should they use?

- a) UTP*
- b) Single-mode fiber*
- c) Multimode fiber*

Your company wants to connect two offices that are about 3 kilometers apart. They want to keep costs down if possible. Which kind of cable should they use?

- a) UTP*
- b) Single-mode fiber*
- c) Multimode fiber*

A switch has the following indication over its network interfaces:



What would happen if you connect it to an identical switch with a straight-through cable?

- a) They would operate normally.
- b) They would operate at a reduced speed.
- c) They would be unable to communicate.

Your company needs to connect many end hosts to a switch which is in a wiring cabinet on the same office floor as the hosts. What kind of cable should they use?

- a) UTP
- b) Single-mode fiber
- c) Multimode fiber

- A) They are connected with a straight-through cable
- C) Multimode fiber
- B) Single-mode fiber
- A) They would operate normally
- A) UTP